



OpenQCM Q⁻¹

QUARTZ CRYSTAL MICROBALANCE
WITH DISSIPATION MONITORING

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openQCM devices are released as scientific open hardware instruments, and they are intended solely for use for **SCIENTIFIC, RESEARCH and DEVELOPMENT APPLICATION, DEMONSTRATION, OR EVALUATION PURPOSES** and are not considered to be finished end-products fit for general consumer use.

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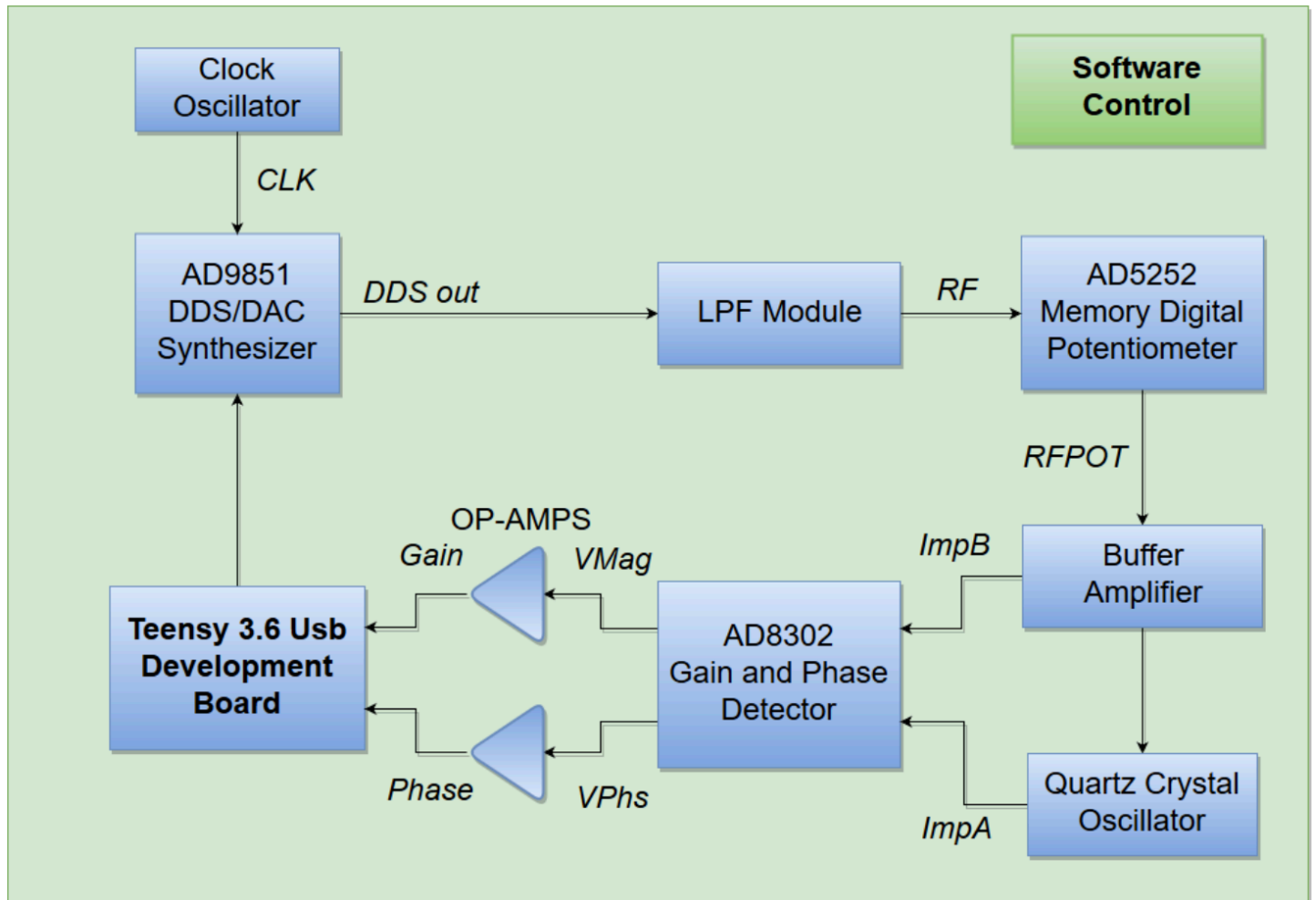
Why openQCM

Although the openQCM are open source systems, they are proper scientific instruments. We are researchers and we firstly build openQCM for our needs. So, we are firmly convinced that high quality research is not necessarily related to highly expensive proprietary products, characterised by closed-architecture.

GENERAL DESCRIPTION

Sensing Principle

openQCM Q-1 is a Quartz Crystal Microbalance instrument capable of measuring simultaneously frequency and dissipation variations. The device is therefore capable of measuring both variations in mass and the viscoelastic properties of the material on the surface of the quartz crystal



The electronics mainly consists of a scalar network analyser, the main block diagram is showed in figure. The scheme of measurement follows the principle of passive interrogation of the quartz sensor by sweeping around the resonance frequency. The actuation signal is generated using the AD9851 DDS/DAC frequency synthesiser and the output signal is read by AD8302 gain and phase detector, which can measure both the magnitude ratio (gain) across the quartz crystal and the phase difference between the actuation and transmitted signal. The analysis of the gain curve allows the characterisation of the sensor by measuring simultaneously the resonance frequency and quality factor. The main advantage of the

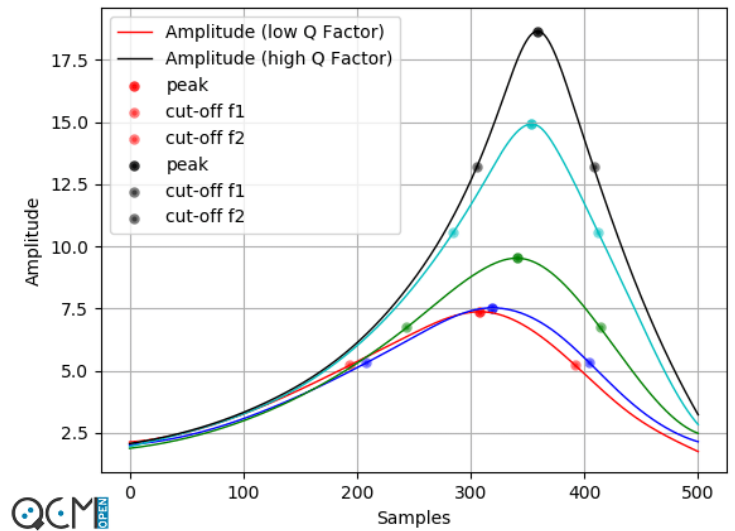
scheme of measurement is the possibility to measure quartz sensor parameters in isolation without an external circuitry influence. openQCM Q-1 electronic consists mainly of two boards: i) the main board features the network analyser and Teensy microcontroller, and ii) the secondary board which is inserted in the fluidic cell, which features the electro-mechanical connections with the quartz crystal sensor using pogo-pin and also a temperature sensor which is mounted as close as possible to the quartz crystal sensor, in order to measure the temperature inside the fluidic cell.

HOW FREQUENCY AND DISSIPATION ARE MEASURED

Each resonance curve is built point by point by sweeping in frequency around the fundamental and overtone harmonics. Once the whole curve is recorded in a buffer, the Frequency is measured by means of a peak detection algorithm. Dissipation is measured by evaluating the quality factor (Q-factor), that is the bandwidth at -3dB for each specific curve.

Q-factor is related to the dissipation D , which is a dimensionless parameter defined as the ratio between the energy loss and stored in each cycle:

Dissipation is an important physical observable because it is related to the viscoelastic properties of the sample in contact with the quartz crystal surface.



ELECTRONICS

Mainboard

The electronics mainly consists of a network analyser that passively interrogates the quartz sensor by sweeping around its resonance frequency.

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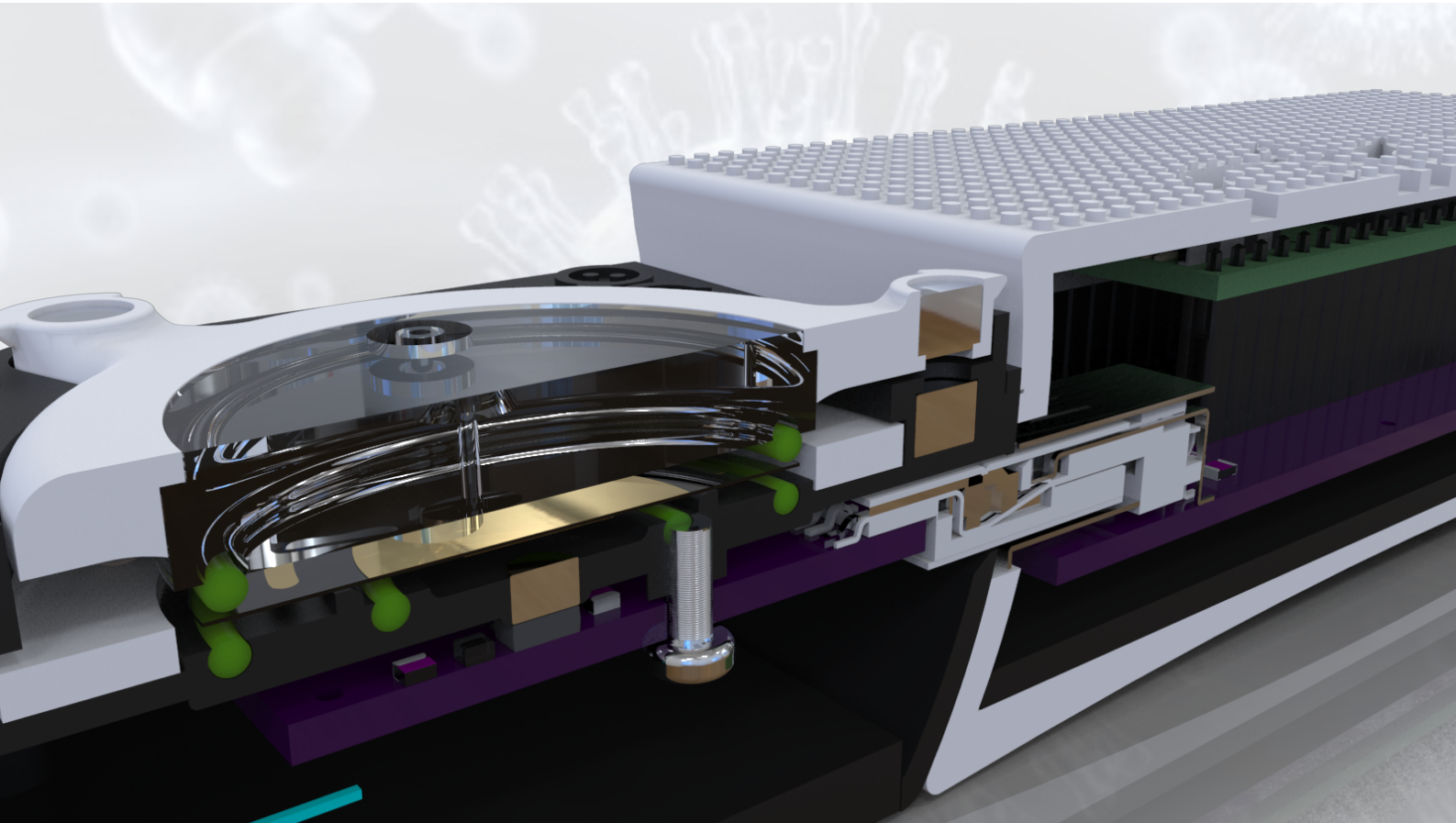
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The mainboard is controlled by a Teensy 3.6 (developed by [Paul Stoffregen](#)), that mounts a ARM Cortex-M4. The board is characterised by a lot of functionalities, the most interesting one is the analog input ADC with an real resolution of 13 bits. It means that over the full-scale range of 3.3 volt it is possible to measure input voltage variations with a resolution 0.4 mV/bits.

HARDWARE

The Design

A new concept design of Quartz Crystal Microbalance

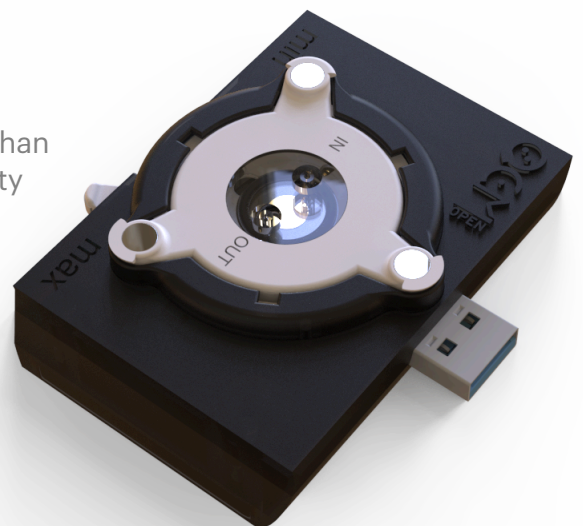


One simple gesture to mount it! The main electronics case is composed of 2 elements produced in 3D printing. With an innovative slot locking system, 2 simple screws are sufficient to ensure compactness and protection for the electronics. An ergonomic design ensures maximum handling for an instrument that was born to be portable.

$$D = \frac{E_{dissipated}}{\pi E_{stored}} = \frac{1}{Q}$$

The new openQCM **sensor module** was born after more than three years of confrontation with the scientific community and is the synthesis of the most important needs we have experienced.

- pogo-pin sensor contacts;
- multi quartz dimension compatibility;
- multi-frequency housing compatibility;
- fine regulation of O-ring; pressure;
- Integrated temperature sensor;
- magnetic back-reaction to sealing.



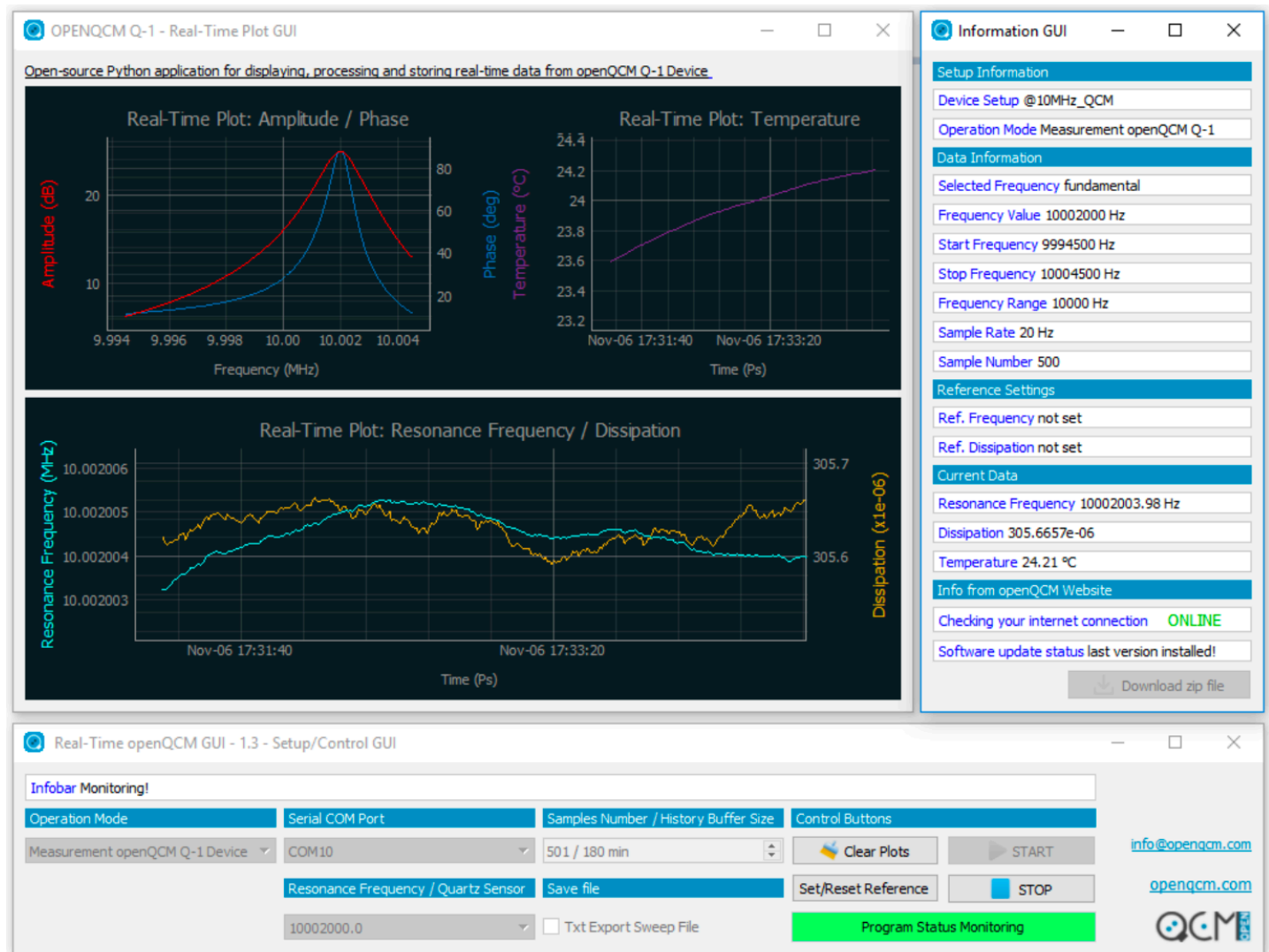
SOFTWARE

GUI Python version 2.1

Minimum requirements

openQCM software is an open-source Python application that displays, processes and stores real-time data from the openQCM Q-1 Device. The application includes internal and external packages needed for using this software.

Intended Audience: Science/Research/Engineering



Name: openQCM Q⁻¹ GUI

Version: 2.1

Programming Language: Python

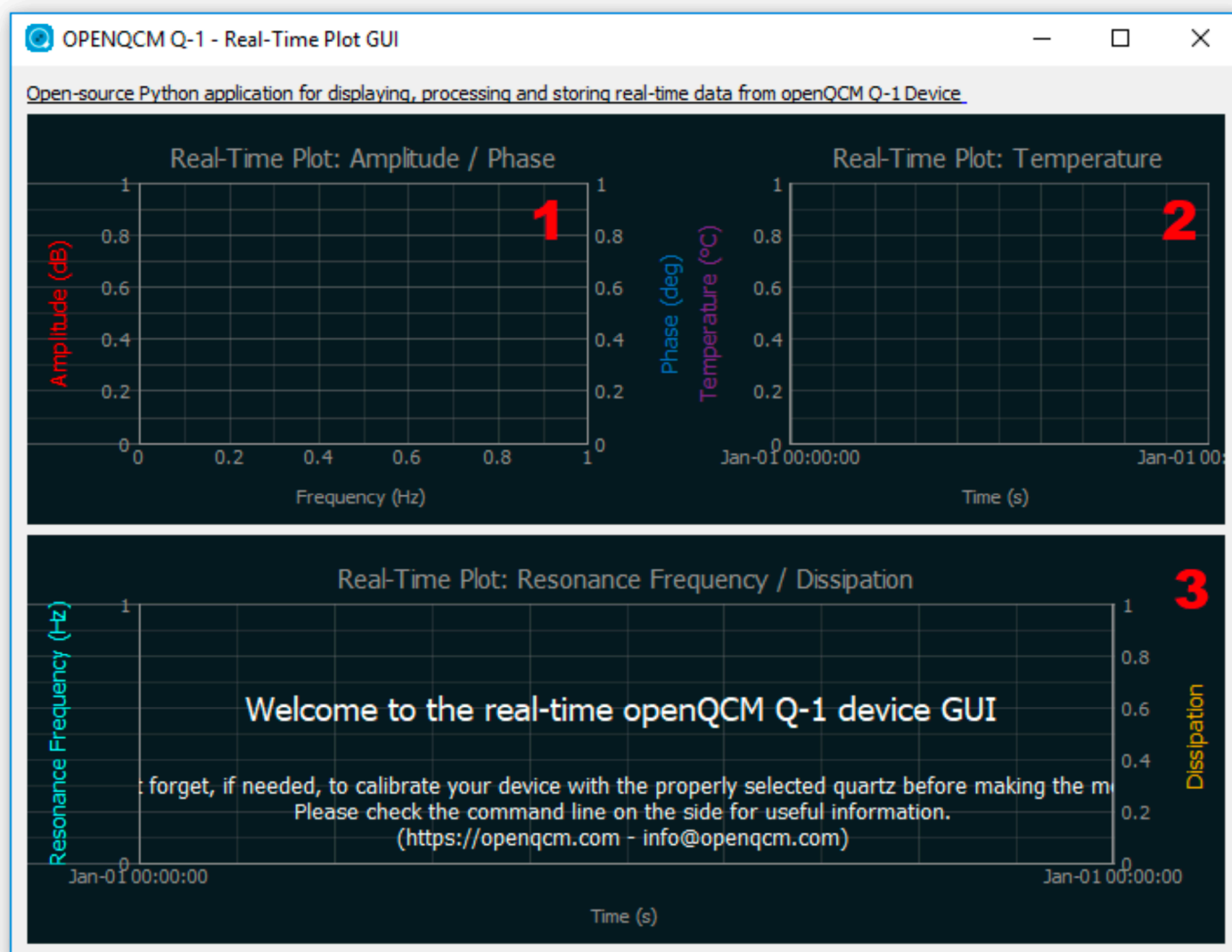
Author: openQCM Team

Supporting and powering the openQCM project: Novaetech S.r.l

SOFTWARE

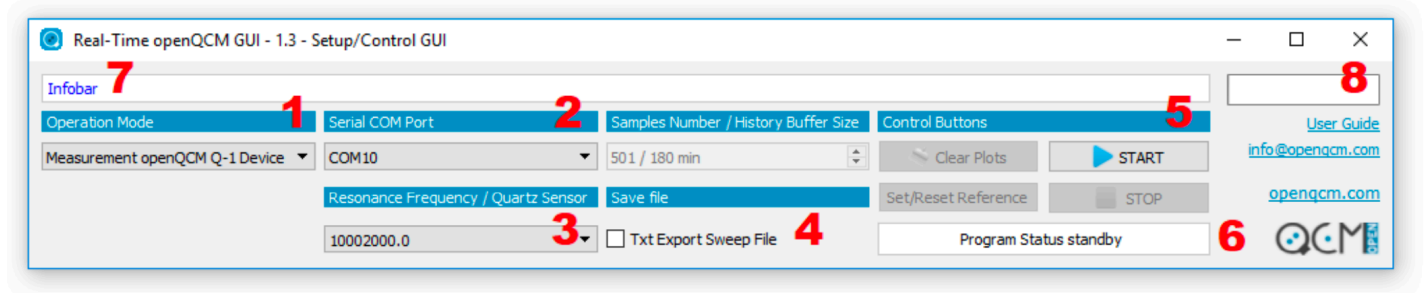
openQCM Q-1 GUI Description

Main interface



The main GUI of openQCM Q-1 Python software version 2.1 is showed in figure. The main GUI consists of three separate windows: 1. 'Setup/Control GUI', 2. 'Real-time Plot GUI' and 3. 'Info GUI'. The detailed description of the functions of each GUI is given below.

1. Setup/Control GUI



Drop-Down Menu Control:

1. Select Operation Mode: "Measurement openQCM Q-1 device" or "Calibration openQCM Q-1 device".
2. "Serial COM Port" selects the available COM port, note that the software shows only the port connected to the device.
3. "Resonance Frequency/Quartz Sensor" drop-down menu: selects the working resonance frequency (fundamental and overtones) in *Measurement openQCM Q-1 device* mode, or select the properly fundamental quartz frequency (i.e. @10MHz or @5MHz) in *Calibration openQCM Q-1 device* mode.

Save File - Radio Button Control:

4. Flag "Txt Export Sweep File" radio button to log data for each sweep cycle in Measurement openQCM Q-1 device operation mode (raw data).

Control Buttons:

5. "START" starts a continuous measurement procedure in Measurement openQCM Q-1 device operation mode OR start the calibration procedure in Calibration openQCM Q-1 device operation mode.

"STOP" stops the acquisition procedure in Measurement openQCM Q-1 device operation mode. Clear Plots: clear all real-time plots (works only in Measurement openQCM Q-1 device operation mode).

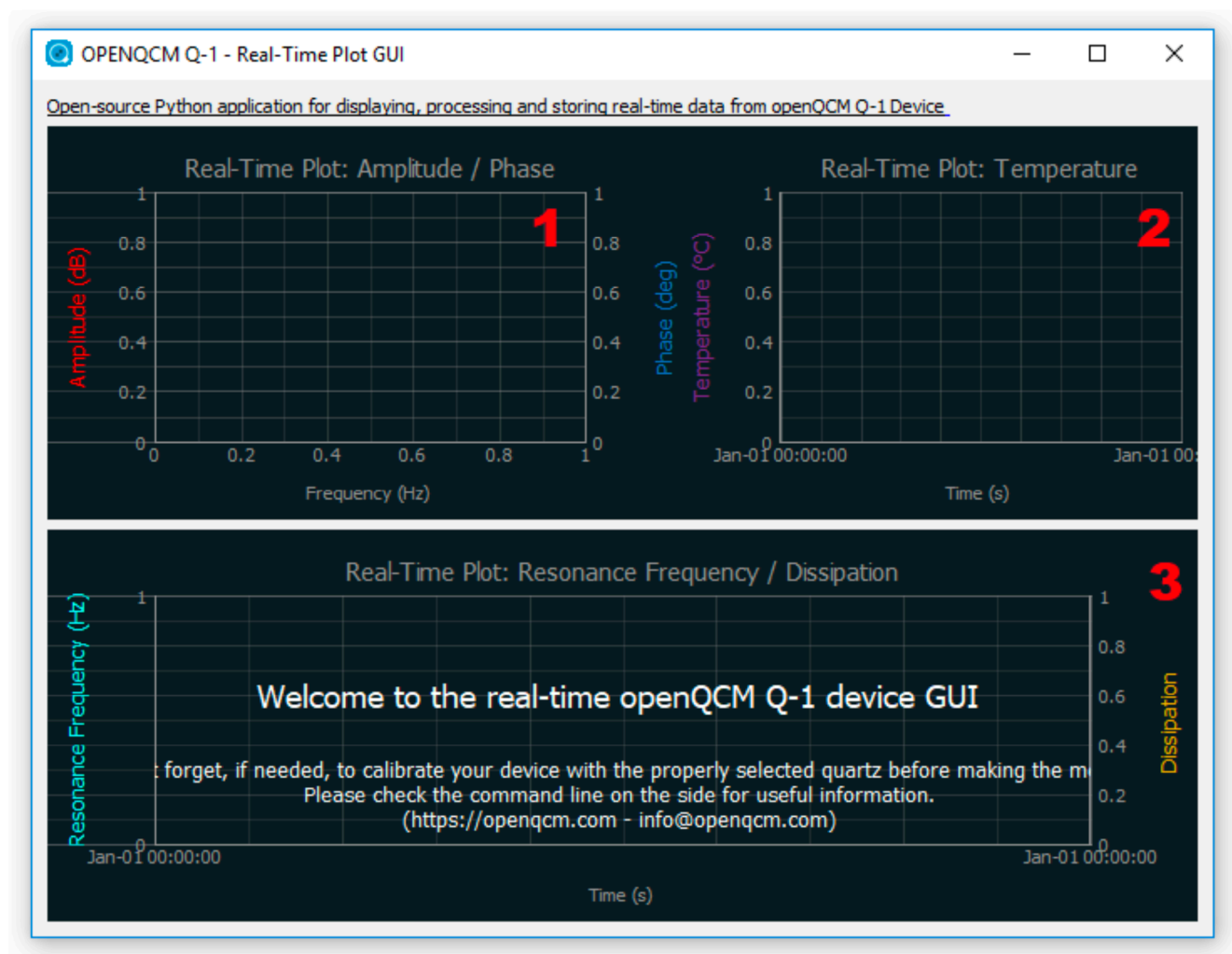
"Clean Plots" cleans all plots in the Real-time Plot GUI (works only in Measurement openQCM Q-1 device operation mode).

"Set/Reset Reference" sets a zero reference line corresponding to the current data: resonance frequency and dissipation respectively (works only in Measurement openQCM Q-1 device operation mode).

Indicators:

6. "Infobar" shows useful additional information, which is shown also in the anaconda prompt.
7. "Program Status" indicator shows the current status of the program and eventually alerts about exceptions (e.g. calibration warnings). Please note that the Program Status indicator changes color (yellow, green and red) depending on the information.
By clicking on the text "openqcm.com" you can access our website or by clicking "info@openqcm.com" you can send us an e-mail. In the latest version of the software you are able to refer to this user's guide online by clicking on the text "User Guide".
8. A progress bar is used to show a user how far along is in a process (works in *Measurement/Calibration* openQCM Q-1 device operation mode).

2. Real-Time Plot

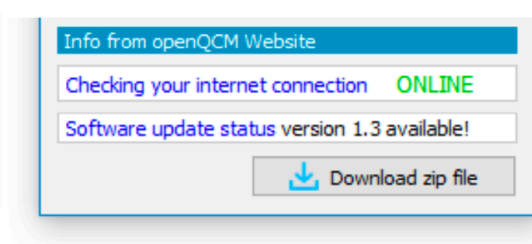
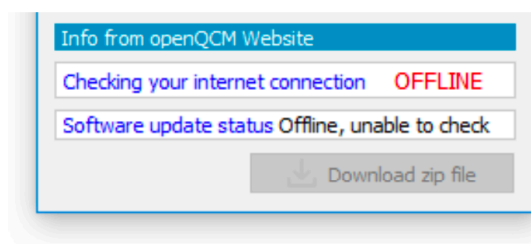
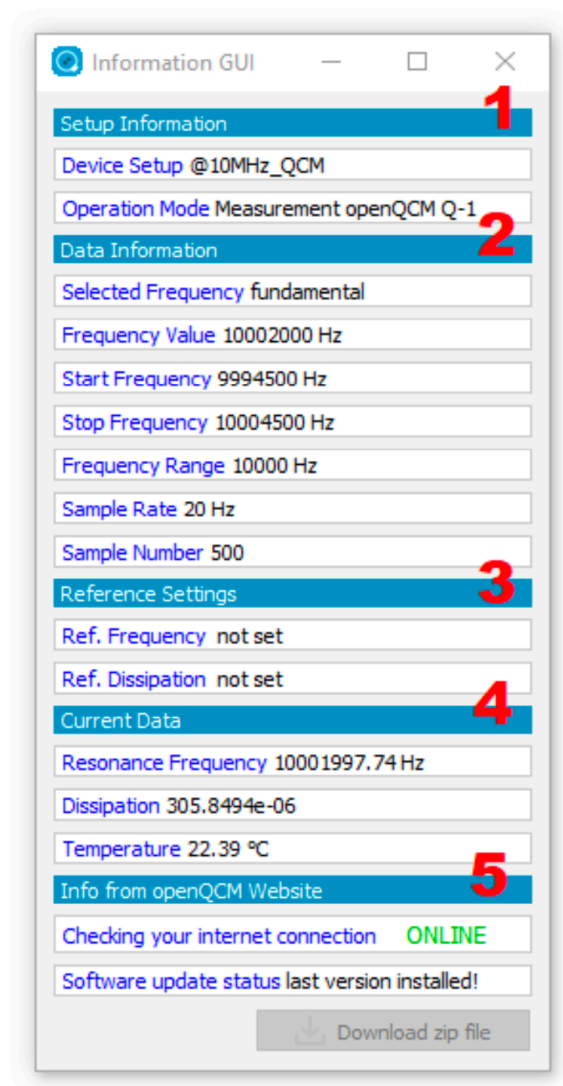


1. "Amplitude /Phase" displays the amplitude and phase curve during both operation modes.

2. "Temperature": displays the current value of the temperature, measured by sensor embedded on the fluidic cell only during *Measurement openQCM Q-1 device* operation mode.
3. "Resonance Frequency/Dissipation" real-time chart simultaneously displays the current frequency and dissipation value, calculated at the end of each sweep in *Measurement openQCM Q-1 device* operation mode.

3. Info GUI

1. "Setup Information" shows the Quartz Sensor and the selected Operation Mode;
2. "Data Information" shows information about selected working resonance frequency
"Reference Settings" indicates the current reference value for resonance frequency and dissipation if "Set/Reset Reference" control button is clicked (works only in *Measurement openQCM Q-1 device* operation mode).
3. "Current Data" shows the current value of frequency, dissipation and temperature in real-time in *Measurement openQCM Q-1 device* operation mode.
4. "Info from openQCM Website" shows if the user is connected to internet and checks if a new version of the software is available. In this case, the "Download zip File" button is enabled.



Example: Internet connection Offline (on the left) and new software version available (on the right)

Techs

Technical Specifications

Sensors and core sensor

Number of sensors	Single quartz resonator sensor
Quartz sensors compatible and tested	5 MHz e 10 MHz , 14 mm blank diameter, - wrapped (single sided contacting)
Volume of measurement chamber	~ 50 μ l

Measurement Specification

Physical quantities	Frequency and Dissipation
Measurement mode	Single overtones (up to 9th overtone for 5 MHz quartz sensors)
Minimum Sampling time	~ 650 ms for each harmonic

Hardware and Material Specification

Main case material	Nylon plastic PA2200
Sensor module materials	Viton (FKM) o-rings, Nylon plastic PA2200, PMMA or PTFE
Main Dimension	(L x W x H): 143.1 x 66.8 x 26 mm
Weight	50 g

Microprocessor Embedded

Teensy 3.6 based on ARM Cortex-M4	
Programming language	C++ arduino - language based code

Software

Real - time frequency and dissipation monitoring and data storage	
Programming language	Python code
OS compatible	Windows, MacOS and Linux OS

Power

Main electronics	USB 5VDC powered (cable included)
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Note: All specifications are subject to change without notice.